

Making Encanto with USD

Rebuilding a Production Pipeline Working from Home

Tad Miller

Walt Disney Animation Studios
Burbank, USA

tad.miller@disneyanimation.com

Harmony M Li

Walt Disney Animation Studios
Burbank, USA

harmony.li@disneyanimation.com

Neelima Karanam

Walt Disney Animation Studios
Burbank, USA

neelima.karanam@disneyanimation.com

Nadim Sinno

Walt Disney Animation Studios
Burbank, USA

nadim.sinno@disneyanimation.com

Todd Scopio

Walt Disney Animation Studios
Burbank, USA

todd.scopio@disneyanimation.com

ABSTRACT

In 2017, Walt Disney Animation Studios began a transition towards Universal Scene Description (USD) as its primary data interchange format [Pixar 2016]. The design and roll out encompassed production on Walt Disney Animation Studios’ “Raya and the Last Dragon” and “Encanto”. In addition to the general challenge of revamping a production pipeline, the studio was fully remote for a majority of the deployment. Even though pipeline changes were extensively tested during development, exercising fundamental pipeline changes in production invariably uncovered many issues. As such, the undertaking required a significant amount of communication and coordination among software developers, technical directors, artists and show leadership.

ACM Reference Format:

Tad Miller, Harmony M Li, Neelima Karanam, Nadim Sinno, and Todd Scopio. 2022. Making Encanto with USD: Rebuilding a Production Pipeline Working from Home. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH ’22 Talks)*, August 07-11, 2022. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3532836.3536236>

1 INTRODUCTION

Deploying a large scale development project, like a new pipeline, into production is a challenge under the best of circumstances. Phased deployment strategies combined with significant testing offer some amount of risk mitigation; however, as the production moves further on its timeline, the cost of change increases dramatically, and stability becomes the utmost priority to allow for maximum artist productivity.

After initial design and development on “Raya and the Last Dragon”, “Encanto” was chosen as the first production to be created natively on the new USD pipeline. However, “Encanto” had a compressed production schedule and was mostly made remotely after the pandemic had started, which added further complexity

to communication and coordination. Despite challenging circumstances, we were able to roll out and create “Encanto” on a new USD pipeline. In this talk, we present our (1) approach to the new pipeline, (2) risk management strategies applied during production, (3) communication channels we used to effectively relay changes to the crew, and (4) takeaways from our experience developing and deploying our new USD pipeline.

2 APPROACH TO THE NEW PIPELINE

2.1 Pipeline Architecture

The USD project initially targeted a smaller subset of consumable data, such as geometry, for replacement with the USD file format. The pipeline was updated to use file translators that read legacy non-USD data formats into USD, so that leaf file formats, scene description formats and user workflows could remain unchanged. In testing, we quickly realized that we would be unable to leverage the power of USD without first updating our core scene description format to USD. As a result, we needed to redesign core pipeline architecture to natively support USD rather than simply consuming USD as a new data source. Consequently, the single team that was initially tasked with the effort of updating software products throughout the pipeline to work with USD needed to grow as the volume of changes grew with decisions to change the core pipeline architecture, and the expertise of other teams was needed to make key architectural decisions.

As the crew grew, the production progressed and our understanding of USD expanded, we found that some of our initial pipeline design decisions did not perform at scale. Frequent crashes and instability caused by incompatibility in our designed workflows and USD’s expectations for data access led us to make changes to accommodate USD’s data access patterns in our underlying version control system. Changes in our asset and shot data structures to USD increased the number of files on disk used for assets and shots and dramatically increased the load on our file servers; this caused us to reevaluate and change some of asset structure and processes to reduce server load.

2.2 Testing

To ensure the successful delivery of multiple significant architectural changes to the pipeline, we chose to maintain workflow parity with the existing toolset in the legacy pipeline. This allowed for a

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH ’22 Talks, August 07-11, 2022, Vancouver, BC, Canada

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9371-3/22/08.

<https://doi.org/10.1145/3532836.3536236>



Figure 1: The house serves as a primary set for "Encanto". This house was created on our new USD pipeline.

clear and common understanding of expected behavior when developing, testing, and rolling out pipeline changes. Targeted test events were held regularly, with participation from technical directors, engineers and artists. Since workflows were largely unchanged, a suite of pipeline tests were developed and refined over time; thus, we were able to quickly generate test events and add focus testing where needed easily.

Targeted artist testing was scheduled into the project development timeline to test changes ahead of release into the main software collection. However, artist testing tended to be more comprehensive when focused on their normal workflow processes than when focused on isolated or targeted testing. Artist testing was subsequently tied to production content generation wherever possible.

2.3 Open Source

We leveraged the open source nature of USD by using existing USD tools as a starting point to help convert some of our workflows. Existing tools in the open source USD ecosystem were helpful in jumpstarting development in some of our workflows. However, as is common when adopting any sizable software package, incorporating these plugins required extensive work to customize and maintain the plugins to fit our studios' production workflows.

3 RISK MANAGEMENT

As with any large scale project that is rolling into an active production environment, risk management is key to minimizing downtime for artists. To reduce risk of destabilizing the pipeline in production, we minimized changes by targeted software testing and releases. Different software collections containing related software changes were available for production testing. Once tested, these collections were rolled into the main software collection for production wide usage. For quick turnaround of software fixes, individual library level releases were made available for artists to locally override and pick up unreleased fixes to continue working.

There were also several instances during production where data on disk needed to be updated to be compatible with the latest software changes. Timing data conversion to reduce impact on artists was very important, especially later in production when significant shot work was happening. Some conversions required artists to

stop and save their work; these conversions were scheduled to happen over the weekend when artists had a natural stopping point to minimize downtime. During conversions, affected software would be released and data would be converted at the same time. Conversion scripts were designed to not only convert the data to the latest structure and format, but also backup the data with the ability to revert and restore to the preconverted state if needed.

4 COMMUNICATION

In addition to the complexities of coordinating test events, data conversions and support for fundamental pipeline changes, a majority of the roll out occurred after the studio started to work remotely. As such, communication was a critical factor to delivering USD on Encanto. Frequent meetings were required for coordination and updates. A regularly scheduled "USD Collective" meeting was created to update production leadership, artistic department supervisors, and studio leadership of progress and upcoming milestones, and to reflect the studio's need to act collectively with a unified direction.

Due to the remote work environment, our messaging system became the main communication hub. Channels for production departments were used to provide daily updates of current issues. Production wide messages were sent for larger updates or for notification when larger issues were encountered.

5 CONCLUSION

The practice of rolling out broad architectural changes during production is a double-edged sword. On one hand, use in production is a forcing function for achieving development milestones, and testing through typical day-to-day production work exercises changes more thoroughly and finds issues faster. However, the constant churn of new features, testing and instability can have both a productivity and a morale cost to production artists and technical directors. This must be taken into account when scheduling and planning large scale changes.

When the production was no longer able to safely absorb additional changes to the pipeline, all but three data sources had been converted to USD. The remaining data stayed in their legacy formats for the duration of the show and was consumed by their native subsystems without translation. Depending on the subsystem involved, this created some fragility in the pipeline as the USD data structure had to mimic the legacy pipeline to a degree for the legacy data format to function. Work to migrate these data formats to USD continued in the studio and has since been completed.

By the time "Encanto" was at the height of shot production, the thoroughness and dedication to design and testing immensely paid off. Through collaboration between technology and production, artists and TDs were able to focus on the process of making the movie on an almost entirely new pipeline based on USD.

REFERENCES

Pixar. 2016. Introduction to USD. <http://openusd.org/>.